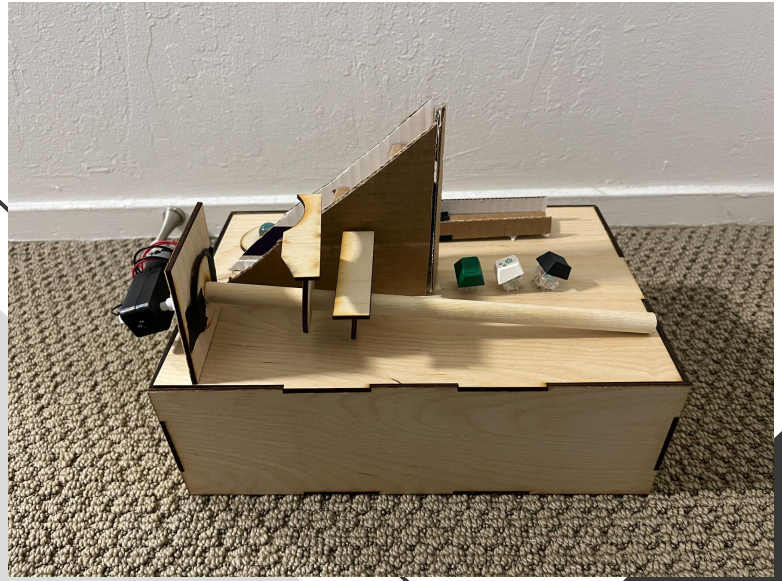


DESCENT

MIDTERM 08B: INITIAL SOUND MACHINE

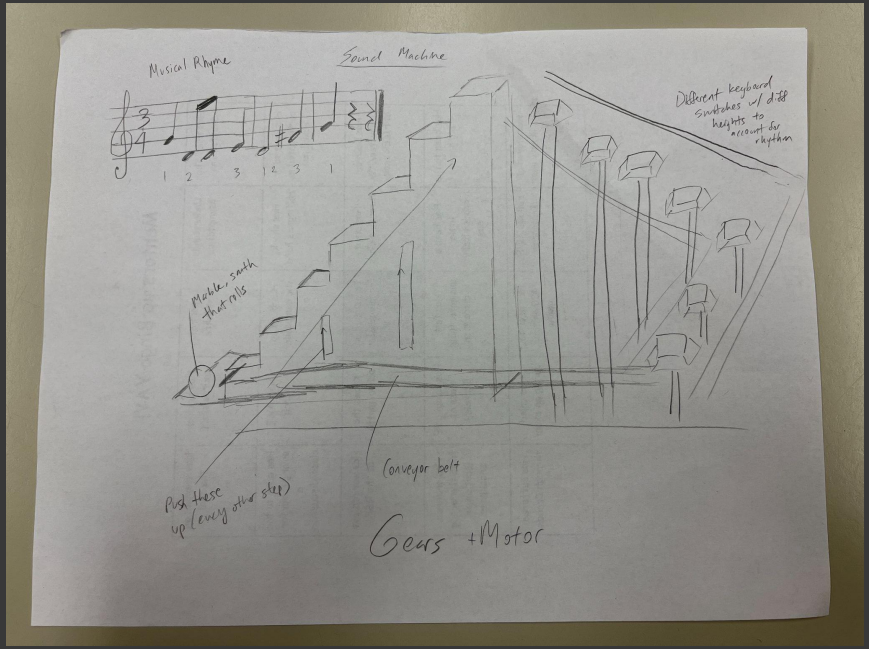
DES INV 22: FALL 2022
10/26/2022

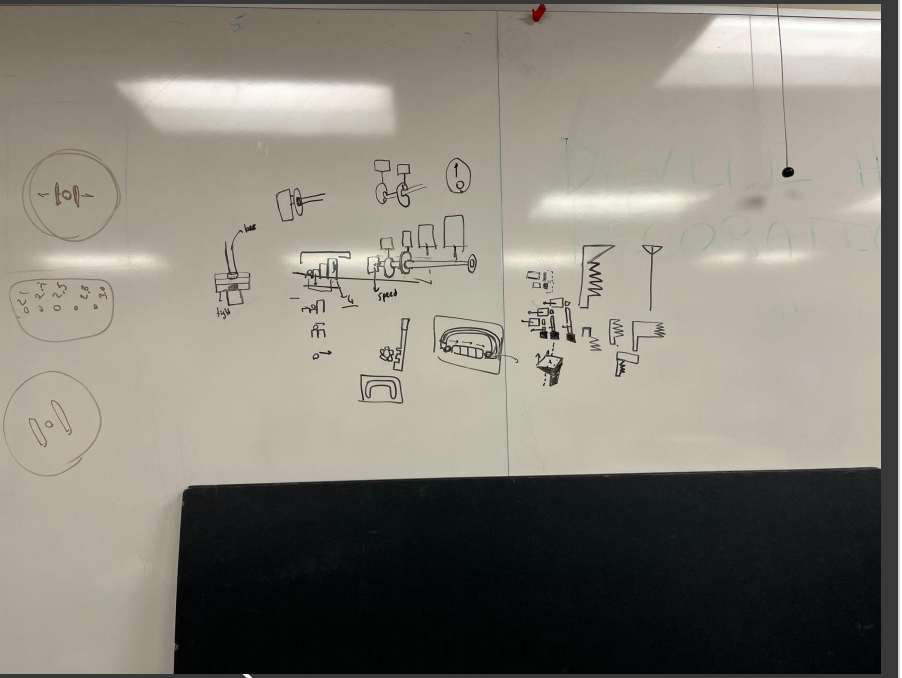
Christy Quang

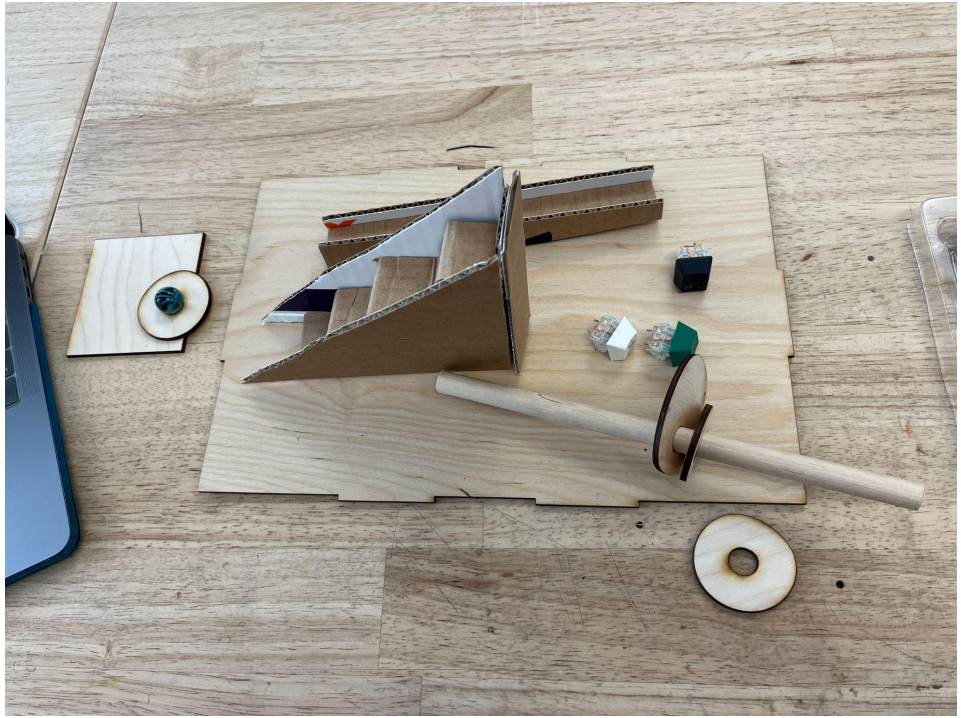
DESCRIPTION

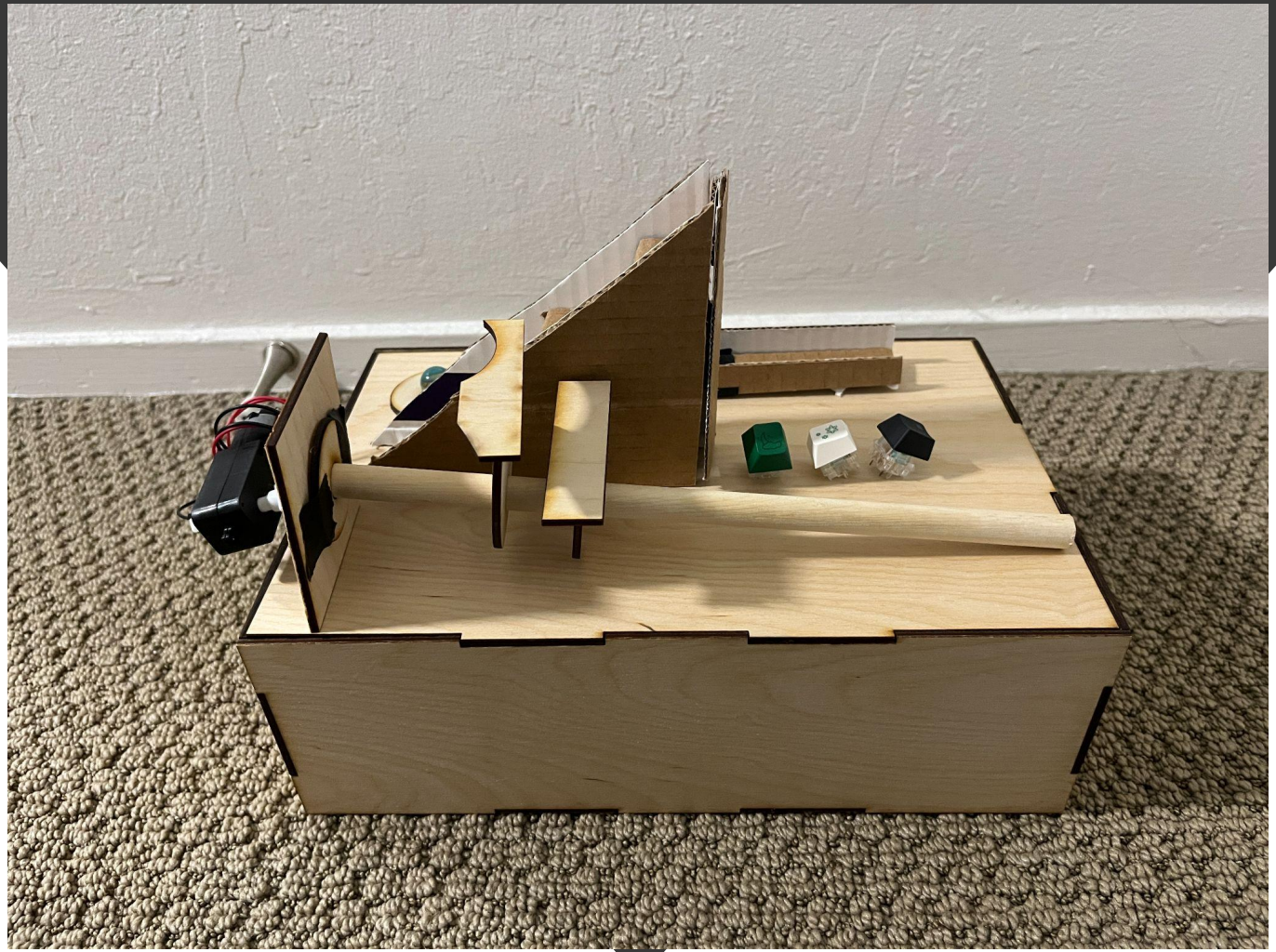
An automatic staircase to make marbles ascend, fall onto keyboard switches to create a musical rhythm, then repeat the process.

Video Link: https://youtu.be/7i535KiBDLQ




base
speed





PROCESS

Deciding between creating a drawing or sound machine was a struggle since both were equally interesting.

Mechanical keyboard switches are loud with a distinct clicking sound which was one of the noises that was desired to be incorporated into the sound machine. Initially, the idea was to hit the different switches with a stick but it was too simple. To increase the complexity of the project, a marble will be brought up a staircase using a motor. At the top of the staircase, gravity will help the marble descend to the ground and during the descent, the marble will hit the different switches, creating different pitches. The different switches and keycaps used will have different pitches which will ultimately create a distinct melody. Finally, the marbles will roll back to the beginning of the staircase where the process will repeat.

To begin prototyping the project, cardboard was used to create a rough outline of the different components in the machine. The box used to store all of the cams, motor, and machine components was laser cut in addition to experimenting with different sizes.

Figuring out how to power the motor will be difficult because the speed needs to be adjusted and set at a slow temperature in order to allow the marbles to roll up each step slowly.

Bill of Materials

- 1 - ½” Wooden dowel
- 1 - Arduino
- 1 - Right Angle Gear Motor
- Battery box
- Breadboard
- Resistor
- 3 - Mechanical keyboard switches & keycaps
- Birch plywood
- Skinny sticks
- Acrylic plastic
- Wood glue
- Handsaw
- Sandpaper
- Paste wax
- Laser cutter
- X-acto knife
- Black Gaffers Tape

Motorized mechanism

My motorized mechanism involves turning a dowel with cams on in order to simulate the steps of a staircase moving up and down, allowing the marbles to ascend each step in a fluid motion. I initially thought that I could use a servo motor to rotate the dowel but that wouldn't be able to rotate 360 degrees.

As a result, I have to use a gear motor with a battery box and resistors to keep the dowel rotating for as long as I want.

REFLECTION

SURPRISES ENCOUNTERED

I thought that reusing the servo motor would in terms of getting my rotation but the limited functionality of that became a surprise encountered. I also thought that I'd be able to connect the gear motor to the battery box but that is unsafe so I'll need to use a breadboard and resistors to make the circuit safe.

FRUSTRATING PARTS

Designing the cams and layout of the stairs was very frustrating. I didn't know how to make the stairs go up and down at different intervals and heights until I had to go to office hours for help. Once I got the concept down, the next frustrating part was getting my initial sound machine to have the correct fitted cam for each individual step.

DIFFERENT APPROACHES

I'm not done with the project yet so I'm not entirely sure on different approaches I would take but so far, I would choose a different central item to focus on. I planned on having the keyboard switches be the final sound but the weight of the marbles aren't heavy enough to individually press the switches so I'll need to buy heavier marbles to account for that.

THANK YOU!